ENGINEERING • ROBOTICS • AUTOMATION

# IPRV V3.1

## Indian Pothole Repair Vehicle

Autonomous detection, localization, navigation and repair




**CAMERA**
surface detection

**DEPTH**
3D geometry

**GPS**
location record

**NAVIGATION**
target positioning

**R5**
material cycle

V3.1 • Prototype technical dossier • Detection  Data  Movement  Repair

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# Project abstract

A 300-word overview of the engineering concept

The Indian Pothole Repair Vehicle (IPRV) V3.1 is a prototype built to address a simple but persistent road-maintenance problem: potholes are usually found late, recorded inconsistently and repaired only after they become a safety issue. The project started from observing damaged road surfaces and asking what information a repair vehicle would need before it could act. Detection alone is not enough; the system also needs to know where the damage is, how large and deep it is, and how to reach the location safely.

IPRV V3.1 combines a four-wheel robotic platform with camera-based computer vision, depth sensing, GPS, autonomous movement and a mechanical repair arrangement. The camera provides the visual input for identifying damaged road areas. Depth sensing adds geometry rather than relying only on a two-dimensional image. GPS links the detected pothole to a physical location, creating a record that can support later maintenance. The navigation system then moves the vehicle toward the target and positions the platform for the repair operation.

The project also follows the R5 waste-management approach: Reduce, Reuse, Retrieve, Redesign and Recycle. Measuring the damaged area before repair can reduce unnecessary material use. Suitable recovered material can be reused or retrieved, while the overall process is redesigned around targeted application instead of repeated manual inspection. Recyclable waste is considered part of the material cycle rather than an automatic disposal step.

V3.1 is therefore not just a moving robot. It is an integrated inspection-and-maintenance concept in which sensing, location, movement and repair are connected in one workflow. The current prototype demonstrates the physical platform and the integration direction; further field testing is required to quantify detection accuracy, navigation error, repair quality, material consumption and reliability under real road conditions. The design aims to reduce worker exposure to traffic and make road-condition data useful for planned maintenance.











# Why the problem matters

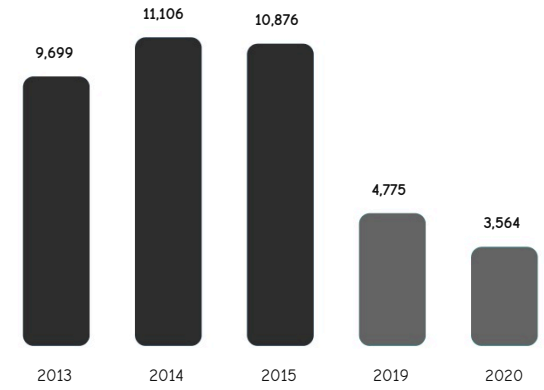
The engineering case begins with road-safety and maintenance evidence

## What makes a pothole difficult for an automated system?

- A pothole is not only a visual defect. Repair decisions depend on location, footprint, depth and the amount of material required.
- Manual inspection can identify damage, but the result is usually a human observation rather than a machine-readable record.
- Repair teams also have to work close to moving traffic. A useful automated platform should separate routine inspection from direct worker exposure where practical.

### Selected reported pothole-related accidents

India • values carried from the project's research chart




Selected years shown; not a continuous time series.

## How the problem was translated into engineering requirements

**01 DETECT**
Find damaged road area from camera input.

**02 MEASURE**
Estimate geometry with depth sensing.

**03 LOCATE**
Attach the observation to GPS coordinates.

**04 POSITION**
Move the vehicle to a useful repair pose.

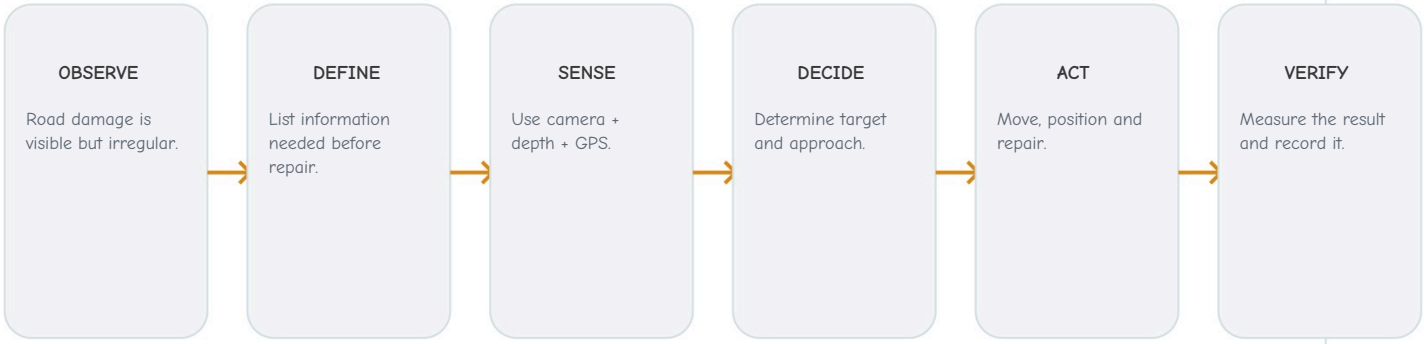
**05 REPAIR**
Apply material through a controlled mechanism.

**06 RECORD**
Keep the observation useful for maintenance planning.

# Problem discovery  system concept

A simple observation was converted into a machine workflow

The problem was not approached as “build a robot and find a use for it.” The starting point was the road itself. Potholes are visually obvious, but the information needed to repair one is not obvious: its position, dimensions, depth, approach path and the amount of material required all matter. A manual repair crew can make these decisions using experience, but a machine needs them to be measured or represented as data.




## Design principle

The key design choice is to connect inspection and repair rather than treating them as separate jobs. The vehicle first creates a description of the defect, then uses that description to guide movement and material use. This reduces the amount of guesswork that would otherwise be pushed onto the operator.




Field photograph of the prototype during outdoor testing.

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# System architecture

Six subsystems form one detect–navigate–repair chain

### 01 ROBOTICS PLATFORM

Four-wheel chassis, geared drive and structural mounting.

### 02 COMPUTER VISION

Camera input used to identify damaged surface regions.

### 03 DEPTH SENSING

Adds distance/geometry information to the visual observation.

### 04 GPS

Associates a detected defect with geographic position.

### 05 AUTONOMOUS NAVIGATION

Converts the target into movement and positioning commands.

### 06 MECHANICAL REPAIR

Handles the physical repair operation after positioning.

## Core data flow

CAMERA → DEPTH → GPS → DECISION → NAVIGATION → REPAIR

The important engineering relationship is not any single sensor; it is the hand-off of information between sensing, decision and actuation.

# Sensing and localization

Turning a road image into a repair target

CAMERA

### Visual input

The camera supplies the surface image used by the detection stage. In a practical deployment, image quality will vary with shadows, water, road texture and lighting.

DEPTH

### Geometry

Depth sensing adds information that a flat image cannot provide reliably. Width, length and depth estimates can become inputs to material estimation and positioning.

GPS

### Location record

GPS links the detected defect to a physical location. That turns one repair into a maintenance record that can be revisited later.




Outdoor prototype view




Front view during controlled testing

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# Autonomous movement and drivetrain

The vehicle has to reach the target before it can repair it

## Movement problem

A pothole detector can produce the correct target and still fail as a repair system if the chassis cannot approach the target consistently. IPRV therefore treats navigation and positioning as a separate engineering stage. The drive system provides the physical movement; the control layer decides where and how the vehicle should move.




| TARGET | APPROACH | ALIGN | STOP | REPAIR |
|---|---|---|---|---|
| GPS / detected point | drive toward target | correct vehicle pose | hold repair position | activate mechanism |

## Drivetrain observations from the prototype

- Large treaded wheels provide useful ground clearance and grip for uneven surfaces.
- Geared DC motors provide the torque needed to move the platform at low speed.
- Motor mounting and wheel alignment are mechanical reliability points because navigation depends on predictable motion.
- The next validation step is to measure commanded distance versus actual distance and quantify stopping/positioning error.

# Repair mechanism and R5

The repair stage is where sensing data becomes physical action

## Repair sequence

| | | |
|---|---|---|
| 01 | POSITION | Vehicle is aligned with the damaged area. |
| 02 | ESTIMATE | Geometry is used to determine the scale of the repair. |
| 03 | DISPENSE | Repair material is supplied through the mechanical system. |
| 04 | LEVEL | Material is distributed across the target area. |
| 05 | VERIFY | Repair quality becomes a test parameter rather than an assumption. |

## R5 applied to the repair process

**REDUCE**

Use geometry to avoid excess material.

**REUSE**

Use suitable recovered material where technically appropriate.

**RETRIEVE**

Recover useful material from damaged sections when feasible.

**REDESIGN**

Shift from repeated manual inspection to targeted intervention.

**RECYCLE**

Keep suitable waste within a material-recovery cycle.

R5 is treated as an engineering constraint: reduce waste first, then consider recovery, reuse and recycling where the material remains suitable.

# Prototype build

Physical construction and field-oriented testing






## Build observations

- Four-wheel layout provides a stable base for the electronics and repair hardware.
- Front-facing sensing is positioned to observe the surface before the vehicle reaches the target.
- The chassis is intentionally compact enough to demonstrate movement in a controlled environment.
- Visible wiring and modular mounting make the prototype easier to modify during development.
- Outdoor photographs show the vehicle being moved and positioned on a paved path and grass-side environment.
- The next build stage should improve enclosure sealing, cable management and service access.
- Mechanical alignment should be checked after repeated movement because wheel position affects navigation.
- Prototype photographs document the current hardware state; they are not performance measurements.

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# Validation plan

What must be measured before claiming field performance

| | | |
|---|---|---|
| DETECTION | **Precision / recall** | Test on labelled road images containing potholes, shadows and wet surfaces. |
| DEPTH | **Absolute depth error** | Compare sensor estimate with a physical measurement of the same defect. |
| GPS | **Position error** | Compare recorded coordinates with a surveyed or reference location. |
| NAVIGATION | **Positioning error** | Measure final vehicle offset from the intended repair pose. |
| REPAIR | **Fill quality** | Measure remaining depression, surface level and repeatability. |
| MATERIAL | **Mass per repair** | Record material loaded, dispensed and recovered. |
| RELIABILITY | **Cycle completion** | Run repeated detect-to-repair cycles and record failures. |

## Acceptance logic

| | |
|---|---|
| Repeatability | The same input should produce similar outputs across repeated runs. |
| Traceability | Each detected defect should retain its location and test record. |
| Safety | The vehicle should stop or fail safely when sensing or navigation becomes uncertain. |
| Material efficiency | Repair output should be related to measured defect geometry rather than a fixed guess. |

# Current limits and next engineering steps

The prototype is a platform for testing, not a finished road machine

### FIELD ROBUSTNESS

Real roads introduce rain, dust, traffic, irregular edges and changing light.

### DETECTION DATASET

Computer vision needs representative labelled examples from Indian road conditions.

### DEPTH CALIBRATION

Depth estimates need physical reference measurements before they can drive material calculations.

### NAVIGATION ERROR

Wheel slip and surface variation can create position error.

### REPAIR QUALITY

A successful dispense cycle does not automatically mean a durable road repair.

### ENCLOSURE

Electronics and sensors need protection against water, dust and vibration.

## V3.1 → V4 development path

01

### CALIBRATE

Camera, depth and GPS reference checks.

02

### BENCH TEST

Repeat detection and motion tests.

03

### CONTROLLED ROAD

Use prepared pothole-like test sections.

04

### FIELD TRIAL

Test under varied real conditions.

05

### MEASURE

Publish actual accuracy, error and material figures.

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# Conclusion

An integrated approach to a practical Indian road-maintenance problem

## What IPRV V3.1 changes

IPRV V3.1 treats pothole repair as an information-and-control problem, not only a mechanical one. The vehicle needs to see the defect, estimate its geometry, know where it is, move to it and then perform a repair operation. Connecting those stages is the main engineering contribution of the prototype.

### SENSE

Camera + depth turn a road defect into usable information.

### MOVE

GPS and autonomous drive connect the information to a physical location.

### REPAIR

The mechanical system converts the selected target into an intervention.

## Final engineering position

The current prototype demonstrates the intended architecture and physical integration. Its strongest next step is measurement: detection accuracy, depth error, GPS error, navigation error, repair quality, material consumption and repeatability should all be recorded with defined test conditions. That evidence will determine how far the system can move from a school-level prototype toward a deployable road-maintenance platform.




IPRV V3.1 prototype in an outdoor test environment.

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# Prototype detail

Selected hardware photographs





